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Code

JupyterLab Python 3 (ipykernel)

[1]:

```
## Part 0 · Warm-Up

print("Hello, BigData!")
```

```
Hello, BigData!
```

[7]:

```
## Part 1 · Variables & Data Types

## Part 1 · Variables & Data Types

my_name = "Aimable"
my_age = 24
my_gpa = 3.4
am_i_present = True

print(my_name ,type(my_name))
print(my_age ,type(my_age))
print(my_gpa ,type(my_gpa))
print(am_i_present ,type(am_i_present))
```

```
Aimable <class 'str'>
24 <class 'int'>
3.4 <class 'float'>
True <class 'bool'>
```

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Code JupyterLab Python 3 (ipykernel)

[3]:
```
##  Exercise 1.2 — Type conversion

age_text = "21"
age = int(age_text)
age = age + 1
print(age)
```

```
22
```

[4]:
```
## Question 1.3 — Think about it

print("3" + "4")
## This will display 34 , Because they are all a string (text) and + operator is going to join instead of adding them as a number
```

```
34
```

[27]:
```
## Part 2 · Strings — Working with Text
## Exercise 2.1 — f-strings

my_name = "Aimable"
my_age = 24

print (f"Hello My name is {my_name}, {my_age} years old, studying at AUCA.")
```

```
Hello My name is Aimable, 24 years old, studying at AUCA.
```

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Code

JupyterLab Python 3 (ipykernel)

```
Hello My name is Aimable, 24 years old, studying at AUCA.
```

[9]:
```
## Exercise 2.2 — String tools

course = "Big Data Analytics"
to_upper = course.upper()
length = len(course)
sub_string = course[0:3]

print (f"Course_Name :{to_upper}")
print (f"lenght :{length}")
print (f"Sub_String :{sub_string}")
```

```
Course_Name :BIG DATA ANALYTICS
lenght :18
Sub_String :Big
```

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Code JupyterLab Python 3 (ipykernel)

```python
scores = [72, 85, 91, 64, 78]
first_number = scores[0]
last_number = scores[-1]
list_length = len(scores)
total = sum(scores)
avg = total / list_length
max_score = max(scores)
min_score = min(scores)

print (f"first_score : {first_number}")
print (f"last_score : {last_number}")
print (f"List_length : {list_length}")
print (f"total : {total}")
print (f" Average : {avg}")
print (f" max_score : {max_score}")
print (f" min_score : {min_score}")
```

```
first_score : 72
last_score : 78
List_length : 5
total : 390
 Average : 78.0
 max_score : 91
 min_score : 64
```

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Code JupyterLab Python 3 (ipykernel)

[5]:
```
## Exercise 3.2 — Lists change

new_scores = scores.append(88)
scores[3] = 66

new_length = len(scores)
new_total = sum(scores)
new_avg = new_total / new_length

print (f"new_scores : {scores}")
print (f"new_avg : {new_avg}")
```

```
new_scores : [72, 85, 91, 66, 78, 88, 88]
new_avg : 81.14285714285714
```

```
## Part 4 · Dictionaries — Labeled Data •••
```

```
Software Engineering
```

```
## Exercise 4.2 — Update and extend •••
```

```
Properties: dict_keys(['name', 'age', 'program', 'district', 'score'])
New_dic: {'name': 'Aimable', 'age': 25, 'program': 'Software Engineering', 'district': 'Kicukiro', 'score': 87}
```

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Code JupyterLab Python 3 (ipykernel)

•[22]:
```
## Part 5 · Class Dataset Analysis

import pandas as pd

df = pd.read_excel("week1_students.xlsx")
df
```

[22]:

| | student_id | name | age | gender | program | district | attendance_pct | score |
|---|---|---|---|---|---|---|---|---|
| 0 | AUCA001 | Aline Uwase | 21 | F | Information Systems | Gasabo | 92 | 85 |
| 1 | AUCA002 | Eric Niyonzima | 23 | M | Software Engineering | Kicukiro | 88 | 78 |
| 2 | AUCA003 | Diane Mukamana | 20 | F | Information Systems | Nyarugenge | 95 | 91 |
| 3 | AUCA004 | Jean Bosco Habimana | 24 | M | Networking | Gasabo | 70 | 64 |
| 4 | AUCA005 | Clarisse Ingabire | 22 | F | Accounting | Musanze | 85 | 72 |
| 5 | AUCA006 | Patrick Mugisha | 21 | M | Software Engineering | Huye | 90 | 83 |
| 6 | AUCA007 | Sandrine Umutoni | 19 | F | Information Systems | Kicukiro | 97 | 94 |
| 7 | AUCA008 | Emmanuel Nsengiyumva | 25 | M | Networking | Rubavu | 65 | 58 |
| 8 | AUCA009 | Grace Uwimana | 20 | F | Accounting | Gasabo | 82 | 76 |

[6]:
```
students = df.to_dict ("records")
students
```

[6]:
```
[{'student_id': 'AUCA001',
  'name': 'Aline Uwase',
  'age': 21,
  'gender': 'F',
  'program': 'Information Systems',
  'district': 'Gasabo',
  'attendance_pct': 92,
  'score': 85},
 {'student_id': 'AUCA002',
  'name': 'Eric Niyonzima',
  'age': 23,
  'gender': 'M',
  'program': 'Software Engineering',
  'district': 'Kicukiro',
  'attendance_pct': 88,
  'score': 78},
 {'student_id': 'AUCA003',
  'name': 'Diane Mukamana',
```

[8]:
```
 ## Exercise 5.1 — Describe the class

scores = [student["score"] for student in students]
```

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Code JupyterLab Python 3 (ipykernel)

[8]:

```
## Exercise 5.1 — Describe the class

scores = [student["score"] for student in students]

print (f"Number of students :{len(students)}")
print (f"Class average :{round(sum(scores)/len(scores)),1}")
print (f"Highest score :{max(scores)}")
print (f"Lowest score :{min(scores)}")
print (f"Score_Range :{max(scores) - min(scores)}")
```

```
Number of students :20
Class average :(77, 1)
Highest score :94
Lowest score :55
Score_Range :39
```

[9]:

```
## Exercise 5.2 — Inspect individual records

print (f"First_student : {students[0]}")
print (f"Last_student : {students[-1]}")
```

```
First_student : {'student_id': 'AUCA001', 'name': 'Aline Uwase', 'age': 21, 'gender': 'F', 'program': 'Information Systems', 'district': 'Gasabo', 'atte
```

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Code JupyterLab Python 3 (ipykernel)

```
Last_student : {'student_id': 'AUCA020', 'name': 'Samuel Rukundo', 'age': 20, 'gender': 'M', 'program': 'Networking', 'district': 'Nyarugenge', 'attenda
nce_pct': 94, 'score': 92}
```

[11]:
```
##  Exercise 5.3 — Count by category

student = students[6]

print(f"{student["name"]} from {student["district"]} studies {student["program"]} and scored {student["score"]}.")
```

```
Sandrine Umutoni from Kicukiro studies Information Systems and scored 94.
```

[14]:
```
## Exercise 5.3 — Count by category

districts = [student["district"] for student in students]
print(districts

print(f"Gasabo : {districts.count("Gasabo")}")
print(f"Kicukiro : {districts.count("Kicukiro")}")
print(f"Gasabo : {districts.count("Nyarugenge")}")
```

```
['Gasabo', 'Kicukiro', 'Nyarugenge', 'Gasabo', 'Musanze', 'Huye', 'Kicukiro', 'Rubavu', 'Gasabo', 'Nyarugenge', 'Muhanga', 'Gasabo', 'Nyamagabe', 'Kicuk
iro', 'Rusizi', 'Musanze', 'Gasabo', 'Nyagatare', 'Kicukiro', 'Nyarugenge']
Gasabo : 5
Kicukiro : 4
Gasabo : 3
```

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Code JupyterLab Python 3 (ipykernel)

[16]:

```
## Exercise 5.4 — Stretch goal: who is the top student?

names = [student["name"] for student in students]

best_score = max(scores)
position = scores.index(best_score)
best_name = names[position]

print(f"The best student is {best_name} with a score of {best_score}.");
```

The best student is Sandrine Umutoni with a score of 94.

[21]:

```
## Bonus

attendances = [student["attendance_pct"] for student in students]

lowest_score = min(scores)
position = scores.index(lowest_score)
lowest_name = names[position]

print(f"The lowest student is {lowest_name} with a score of {lowest_score} and attendance of {attendances[position]}%.")
```

The lowest student is David Hakizimana with a score of 55 and attendance of 72%.

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Code JupyterLab Python 3 (ipykernel)

```
ndance_pct': 92, 'score': 85}
Last_student : {'student_id': 'AUCA020', 'name': 'Samuel Rukundo', 'age': 20, 'gender': 'M', 'program': 'Networking', 'district': 'Nyarugenge', 'attenda
nce_pct': 94, 'score': 92}
```

[11]:

```python
##  Exercise 5.3 – Count by category

student = students[6]

print(f"{student["name"]} from {student["district"]} studies {student["program"]} and scored {student["score"]}.")
```

```
Sandrine Umutoni from Kicukiro studies Information Systems and scored 94.
```

[14]:

```python
## Exercise 5.3 – Count by category

districts = [student["district"] for student in students]
print(districts

print(f"Gasabo : {districts.count("Gasabo")}")
print(f"Kicukiro : {districts.count("Kicukiro")}")
print(f"Gasabo : {districts.count("Nyarugenge")}")
```

```
['Gasabo', 'Kicukiro', 'Nyarugenge', 'Gasabo', 'Musanze', 'Huye', 'Kicukiro', 'Rubavu', 'Gasabo', 'Nyarugenge', 'Muhanga', 'Gasabo', 'Nyamagabe', 'Kicuk
iro', 'Rusizi', 'Musanze', 'Gasabo', 'Nyagatare', 'Kicukiro', 'Nyarugenge']
Gasabo : 5
Kicukiro : 4
Gasabo : 3
```

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Code

JupyterLab Python 3 (ipykernel)

```
The lowest student is David Hakizimana with a score of 55 and attendance of 72%.
```

[ ]:
```
## Part 6 · Reflection


# Q1. Is this 20-student dataset Big Data?

    # Ans:  No. It is a small dataset, not Big Data. It has only 20 student records, so it is easy to store and analyze.
    #   The 5 V's of Big Data are Volume, Velocity, Variety, Veracity, and Value. This dataset clearly does not have enough Volume because there are onl

# Q2. Something that surprised or confused me:

      # Ans: One thing that surprised me was that Python treats numbers and text differently. For example, "3" + "4" gives "34" instead of 7 because the
```

[ ]: